

Since there is full employment in the economy all the time, with a given production function and capital, the Aggregate Supply (AS) is inelastic at full employment level of output. In Fig 1.3 we have shown the AS curve as a vertical straight line. Note that the AS curve will shift if there is a change in the level of technology (production function) or the level of capital.

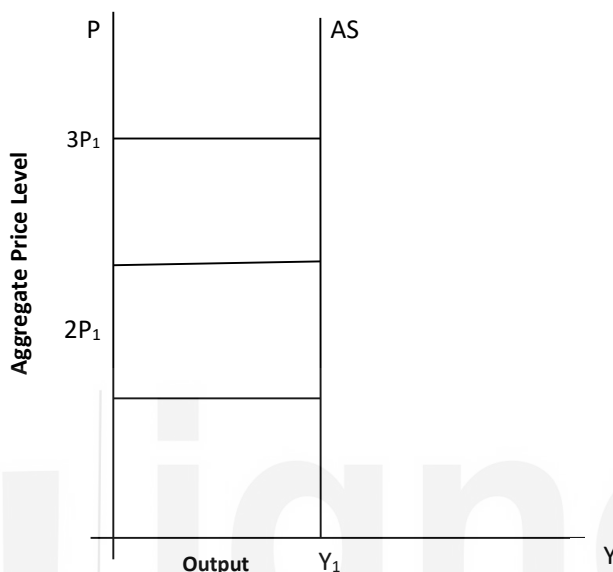


Fig. 1.3: Aggregate Supply Curve

1.5 QUANTITY THEORY OF MONEY

To understand the effects of changes in the stock of money, we need to study the money market equilibrium. Money is a stock variable. Its stock means its quantity at a point of time. Money is an asset demanded by the public for its holding. Money is supplied by the government and the banking system. The interaction of the demand for money and the supply of money create a money market. In this unit, we assume that the monetary authority autonomously determines the supply of money in an economy. Hence, the aggregate demand for money refers to the demand for money by the public. Thus, demand for money means the sum of all the money demanded by individual members of the public, whether households or firms.

There are various theories of demand for money, such as (i) the classical theory of demand for money, or the Quantity Theory of Money (QTM); (ii) the Keynesian theory of demand for money; and (iii) Friedman's restatement of Classical QTM. In this Unit, we will discuss the classical theory of demand for money. It is popularly known as the QTM, which is basically a theory of the price level. Recall that money has four functions; (i) a medium of exchange, (ii) a standard of deferred payment, (iii) a store of value, and (iv) a unit of account. According to classical economists, money is demanded for its function as a medium of exchange. There are several versions of the popular classical QTM. Below we present the transaction version, which is known as the Fisher's equation of exchange:

$$M.V = P.T \quad \dots (1.5)$$

where,

T is number of transactions of average size, and proxy for income level

M is quantity of money supply,

V is velocity of circulation of money, and

P is the average price level.

Classical economists rely heavily on the QTM to establish the theory of demand for money. The QTM says that there is a proportional relationship between the amount of money held by the public and the price level. As per the classical assumption of full employment, the output level is given. Hence, T in equation (1.5) is fixed as a proxy for national income. Further, V is defined as the number of times a rupee moves from one hand to another during a given period. This means that it relies on the public's payment behaviour, which is constant in the long run.

The right-hand side of equation (1.5), PT represents the total amount of money required to facilitate transactions, i.e., the purchase or sale of total output in the economy. The left-hand side of equation (1.5) In the same equation, the left-side term MV is the product of the number of rupees in circulation and the number of times each is used for public payments. Hence, MV represents the total amount of money available for transactions during the given period. Equilibrium is achieved in the system at the point where money demanded (PT) equals money supply (MV).

The equation (1.5) can be rewritten as below,

$$P = \frac{V}{T} M \quad \dots (1.6)$$

As mentioned earlier, the terms V and T are constants in equation (1.6). Thus, the equation indicates that money supply directly affects price level. For instance, if M is increased by four times, price level P will rise by four times. Therefore, classical QTM is known as a theory of price level.

The other approach of classical QTM describes the relationship between the demand for money and 'nominal output' by taking directly real output instead of number of transactions. In equation form, it can be expressed as

$$MV = PY \quad \dots (1.7)$$

where,

M is the money supply

V is the velocity of money that is the number of times the money changes hand.

P is the output price, and

Y is the real output level.

The total stock of money in the economy, measured by money supply times the velocity of circulation (MV), equals the nominal value of transactions or the nominal value of income or output in the economy (PY). The identity given at (1.7) is converted into the QTM under the assumption that Y and V are stable or constant in the short run. With Y and V being constant, the assumption that price level is passive means that P depends on changes in M rather than M depends on changes in P . These assumptions indicate that any short-run decrease (or increase) in M must lead to a proportional decrease (or increase) in P . Relaxation or invalidity of any one of these assumptions would not hold the proportionality relationship of P with M .

According to classical theory, under perfect competition with full employment in the economy, the output level is fixed ($\bar{Y}$). The velocity of circulation is predetermined ($\bar{V}$) and money supply is exogenously given. From equation (1.6), Price level (P) is proportional to the money supply (M).

$$P = (\bar{V}/\bar{Y}) \times M \quad \dots (1.8)$$

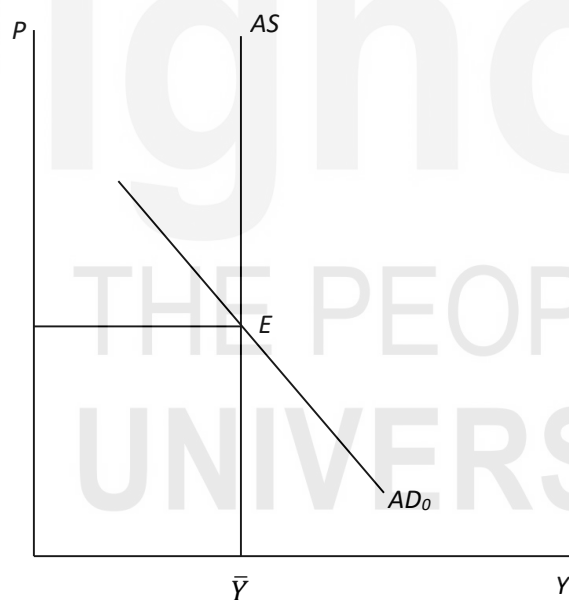


Fig. 1.4: Equilibrium Output and Prices

Thus, the labour market and production function determine the output level while money supply determines the price level in the classical scheme. The quantity theory of the money also gives the aggregate demand curve (see Fig. 1.4). Although the classical theory is accepted by most economists, Keynesian economists and Monetarists have criticized it. They believe that the classical theory fails in the short run when the prices are sticky. Also, it has been established that the velocity of money does not remain constant over

time. Nevertheless, the classical theory can be used to understand and control inflation in the economy.